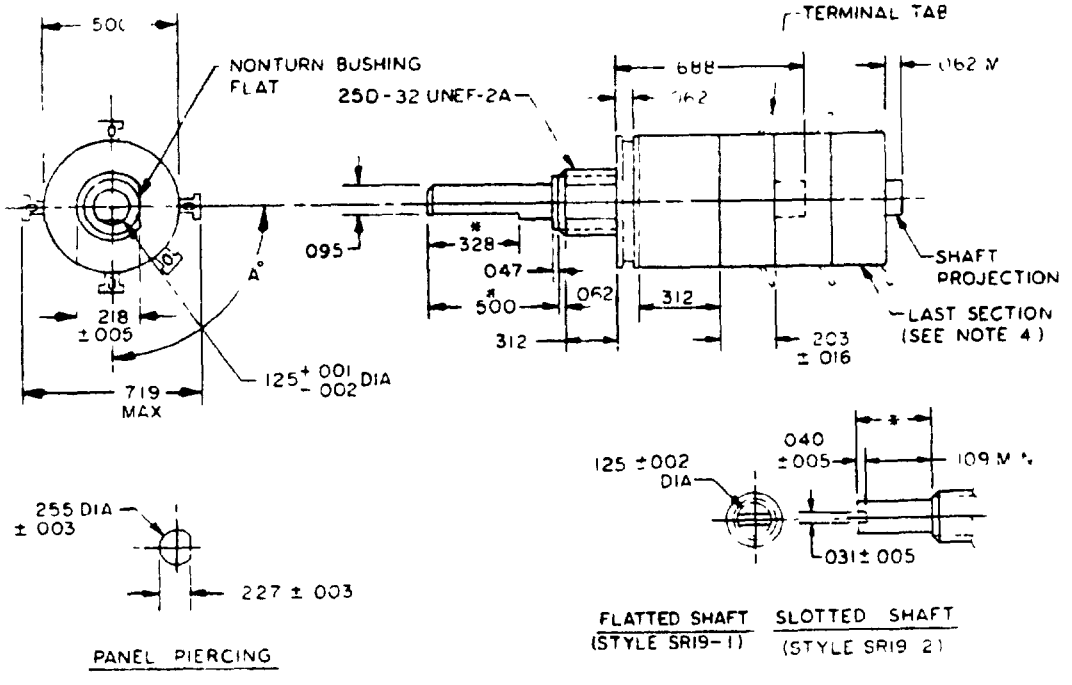
MILITARY SPECIFICATION SHEET

SWITCH, ROTARY, OPEN CONSTRUCTION, 1/10 AMPERE,
STYLE SR19

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-S-3786.

This specification is approved for use by all Departments and Agencies of the Department of Defense.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.031	.79	.125	3.18	.255	6.45
.002	.05	.040	1.02	.203	5.16	.312	7.94
.003	.08	.047	1.19	.218	5.54	.328	8.33
.005	.13	.062	1.59	.227	5.77	.500	12.70
.016	.41	.095	2.41	.250	6.35	.686	17.48
		.109	2.78			.719	18.26

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.015 (0.38 mm).
4. The number of sections is optional. Maximum 5 sections.
5. A dimension angle on flat of shaft.
6. For D&F construction switches the bushing shall be flatted to accommodate the panel cutout as shown. Nonturn device as shown is not applicable to D&S construction switches
7. For allowable variations, see MIL-S-3786 ordering data.
8. Shaft-flat angle A° is the angle between a line through the center of the shaft and center of the nonturn device and another line through the center of the shaft and perpendicular to the shaft flat.
9. Shaft shown in maximum counterclockwise position for switches with stop and with switch in position No. 1 for switches without stops.

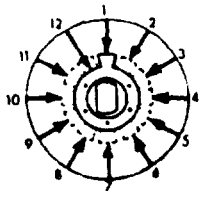
FIGURE 1. Style SR19 switch.

FSC 5930

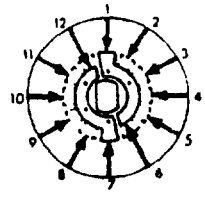
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1/2" SWITCHES STANDARD CIRCUIT CONFIGURATIONS

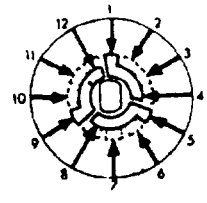
NOTE All layouts are viewed from knob end with switch in extreme counter clockwise position.



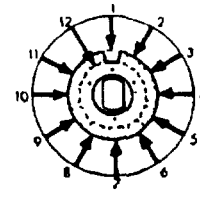
1 POLE
11 POSITION
30° THROW



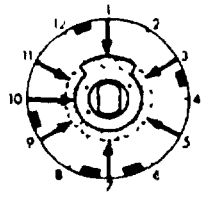
2 POLE
5 POSITION
30° THROW



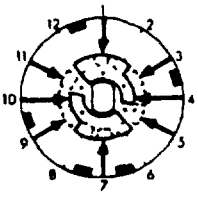
3 POLE
3 POSITION
30° THROW



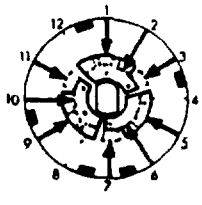
NOTCHED BLADE
11 POSITION
30° THROW



1 POLE
6 POSITION
60° THROW



2 POLE
3 POSITION
60° THROW



3 POLE
2 POSITION
60° THROW

FIGURE 2. Circuit diagrams viewed with switch in extreme counterclockwise position

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REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Angles of throw: 30° and 60° (see table IV).

Terminals: Terminals shall be designed so that the terminals can accommodate a wire having a diameter of 0.020 inch. The terminal shall be bent at an angle of 60° +30°/-15° from the plane of the section.

Rotational torque: The minimum and maximum values of torque determined for shaft rotation shall be within the limits specified in table I.

TABLE I. Rotational-torque limits.

Temperature	Torque (lb-in)	
	Minimum	Maximum
Room	0.5	2.0
Minimum	0.7	2.2

Mounting hardware: Each switch shall be supplied with one each hexagon nut in accordance with MS25082-C13 or its equivalent and one internal-tooth lock washer in accordance with MS35333-74 or its equivalent.

Construction: B, D, E, or F (see table III).

Terminal strength (pull): A force of 1 pound shall be applied to the terminals.

Stop strength: Stop shall withstand a force of 5 pound-inches.

Temperature-life characteristic: Temperature-life characteristic shall be 10,000 cycles at -65°C and +85°C (see table III).

Altitude: Symbol C (70,000 ft).

Vibration grade: Symbol 3 (10 to 2,000 Hz).

Insulation material: Symbol P (see table III).

Shock type: Symbol H (high impact), and symbol M (medium impact) (see table III).

Life (rotational): The test loads for the applicable circuit conditions shall be as specified in table II. Each of the loads specified for the applicable environmental condition shall be switched by at least one rotor contact of the switch.

TABLE II. Electrical loads.

Environmental condition	Loads			
	Inductive (2.8 henries)		Resistive	
	Milliamperes	Volts, dc	Milliamperes	Volts, dc
At atmospheric pressure	10	28	100 25	28 115 V rms
At reduced barometric pressure	---	---	50	28

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Dielectric withstanding voltage. The magnitude of test voltage for the dielectric withstanding voltage test shall be 500 V ac at atmospheric pressure, and 100 V ac at reduced barometric pressure.

Corona: The corona test potential shall be 500 volts.

Contact resistance: The contact resistance at room temperature shall not exceed the following values:

Initial and after vibration and shock - - - - - - - -	20 milliohms.
After moisture resistance and salt spray - - - - - - -	30 milliohms.
After initial soak and completion of life-test - - -	40 milliohms.

Part number: The military part number shall consist of M3786/19- (selected from tables III through V) as shown in the follow example (see note):

M3786/19- A B A S

- A: Construction type, temperature-life characteristic, shock type, and insulation material (see table III).
- B: Poles per deck, number of positions per pole, and angle of throw (see table IV).
- A: Number of decks and switching characteristics (see table V).
- S: Slotted or flatted shaft (see table VI).

NOTE: M3786/19-ABA identifies a rotary switch of construction type B, temperature-life characteristic A, shock type H, insulation material plastic; 30° angle of throw, one pole per deck, 2 positions; one deck and a nonshorting switching characteristic.

TABLE III. Code letters for combinations of construction type, temperature-life characteristic, shock type, and insulation material.

Code letter	Construction type	Temperature-life characteristic	Shock type	Insulation material
A	B	A	H	P
B	D	"	"	"
C	E	"	"	"
D	F	"	"	"
E	B	"	M	"
F	D	"	"	"
G	E	"	"	"
H	F	"	"	"

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TABLE IV. Code letters for combinations of poles per deck, number of positions per pole, and angle of throw.

Code letter	Poles per deck	Number of positions	Angle of throw
A	1	2	30°
B	"	3	"
C	"	4	"
D	"	5	"
E	"	6	"
F	"	7	"
G	"	8	"
H	"	9	"
J	"	10	"
K	"	11	"
L	"	12 1/	"
M	2	2	"
N	"	3	"
P	"	4	"
Q	"	5	"
R	3	2	"
S	"	3	"
T	1	2	60°
U	"	3	"
V	"	4	"
W	"	5	"
X	"	6 1/	"
Y	2	2	"
Z	"	3	"
1	3	2	"

1/ Switch is continuous rotation type (no stops), and number of positions is dependent on angle of throw as follows:

Angle of throw	Positions
30°	12
60°	6

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TABLE V. Code letters for combinations of number of decks and switching characteristics.

Code number	Number of decks	First deck	Second deck	Third deck	Fourth deck	Fifth deck
A	1	NS				
B	1	S				
C	2	NS	NS			
D	2	S	S			
E	2	NS	S			
F	3	NS	NS	NS		
G	3	S	S	S		
H	3	NS	NS	S		
J	3	NS	S	S		
K	4	NS	NS	NS	NS	
L	4	S	S	S	S	
M	4	NS	NS	NS	S	
N	4	NS	S	S	S	
P	4	NS	NS	S	S	
Q	5	NS	NS	NS	NS	NS
R	"	S	S	S	S	S
S	"	NS	NS	NS	NS	S
T	"	NS	S	S	S	S
U	"	NS	NS	NS	S	S
V	"	NS	NS	S	S	S

TABLE VI. Code letter configuration for slotted and unslotted shaft.

S	Slotted shaft	(SR19-2)
F	Flatted shaft	(SR19-1)

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AT, AV, SM
Navy - MC, CG
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Air Force - 85
(Project 5930-1243-8)

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